Distilling Runtime Safety into VLA Robot Policies

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Abstract

Vision-language-action (VLA) models generalise across manipulation tasks but carry no notion of physical safety: on the SafeLIBERO benchmark, $\pi_{0.5}$ displaces a non-target obstacle in 82.5% of episodes. The standard remedy is a runtime control-barrier-function shield, which projects each proposed action onto a safe set. It works, but it is permanent — consuming inference budget at every control step, requiring obstacle geometry at deployment, and adding a component that can itself fail.

This thesis asks whether that corrective behaviour can instead be amortised into the policy's weights. Distilling $\pi_{0.5}$ on its own shielded rollouts reduced collisions to 19.2% with no shield at inference, while raising task success from 58.3% to 82.5% on held-out initial states. One distillation round recovers most of the shielded baseline's collision avoidance (80.8% against 86.7%), and the shield finds materially less to correct when reattached, firing on 18% of control steps against 36%.

A geometry-privileged motion planner, supplying corrections of equal density under the same shield, produced no measurable improvement under a matched comparison; filtering for successful episodes alone produced none either. What transfers is correction at states the policy itself reaches.

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Chapter 1

Introduction

Vision-Language-Action (VLA) models introduce a paradigm shift in the field of robotics, serving as a primary driver towards general-purpose embodied intelligence. By extending the capabilities of Vision-Language Models (VLMs) into physical execution, VLAs unify multimodal perception and low-level control into single end-to-end architectures, replacing classical modular pipelines.

However, deploying VLA policies into real-world, human-centric environments introduces severe safety risks. While hallucinations in pure language models merely yield incorrect text, ungrounded actions or spatial errors in VLAs directly cause physical collisions, property damage, or catastrophic injury. The aim of this work is to determine whether the safety gains provided by external inference-time mechanisms can be in turn internalised in the model weights, and thus making the VLA inherently safer. This chapter starts by providing background context, before presenting the research problem, rationale, scope, and significance of the study.

1.1 Background

Robot control was until recently built from modular pipelines. VLA models collapse these pipelines into a single network trained end-to-end, mapping camera images and a natural-language instruction directly to motor commands [2]. Pretraining on web-scale vision-language data and large cross-embodiment robot datasets has made these models strikingly general. [11].

Generality does not imply safety, however. A VLA is trained to reproduce demonstrated behaviour, so nothing in its objective penalises contact with an object that is not the target, and nothing in the data reliably demonstrates avoidance as human demonstrators rarely produce the near-collisions that would teach it. The consequence is measurable: on the benchmark used throughout this work, an unmodified $\pi_{0.5}$ disturbs a non-target obstacle in more than eight of every ten episodes. Throughout this thesis, *safety* refers narrowly to avoiding unintended physical contact with objects in the workspace, rather than to the broader alignment concerns associated with language models.

Nevertheless, safety has not gone unaddressed, and the approaches fall into three main clusters. The most prominent places a corrective component between the policy and the robot, checking each proposed action against an analytic or learned safety condition and modifying it where the condition is violated [8, 10, 14]. A second line of work changes what the policy optimises, expressing safety as a constraint on the learning problem rather than as a filter on its output [15, 16]. A third distils a corrective signal into the policy's weights, so that safer actions are produced rather than intercepted [17]. Purpose-built evaluation has followed: SafeLIBERO augments an established manipulation suite [9] with obstacles placed deliberately in the robot's path [8], and is the setting for everything reported here.

A control barrier function defines a safe region of the state space and derives, from it, a condition on the robot's velocity that keeps the system inside that region: satisfy the condition at every instant, and a system that starts safe provably remains safe [1]. Because that condition is linear in the commanded velocity, it can be enforced by a small optimisation solved at each control step, which takes whatever action a controller proposes and returns the nearest action that satisfies the constraint. Such a component is commonly called a *shield* or *safety filter*: it sits between the policy and the robot, passing safe actions through unchanged and minimally correcting unsafe ones.

Applying this machinery to VLA models is recent. The immediate prior work inserts such a filter between $\pi_{0.5}$ and the robot, grounding obstacle geometry from the model's own observations, and demonstrates a large reduction in collisions while preserving the policy's ability to follow instructions [8]. Runtime shielding is therefore the current state of the art for making a VLA safe and is the starting point for the problem this thesis addresses.

1.2 Problem Statement

That a learned policy can be made safer by an external component is established. A filter substantially reduces collisions on SafeLIBERO while preserving instruction-following [8], and learned variants achieve similar ends without an analytic barrier [10, 14]. Separately, safety-relevant behaviour has been shown to transfer into a policy's weights: by imitating a robust barrier-based expert under stated conditions [4], by collecting corrections at the states a learner's errors produce [3, 7], and by distilling a corrective signal into a VLA policy directly [17].

What has not been established is whether the two approaches can be joined. Every filtering method retains its corrective component at deployment, solving an optimisation or running a gradient correction at each control step. Every transfer result either requires conditions no large visuomotor policy satisfies [4], or distils from a teacher constructed out of the student itself [17]. Whether the corrections a control-theoretic shield produces can be compiled into a policy's weights, such that the shield can be removed, has not been explored.

The gap matters because the filter is a mitigation rather than a solution, and because its benefit and its cost arise from the same mechanism. It is imperfect: even in the strongest published configuration on this benchmark, roughly a quarter of episodes still end with an obstacle displaced [8]. And it interferes — the authors of that work report that enforcing a constraint can drive a policy into states its training never covered, where it may behave erratically and fail to recover [8]. Correcting actions at runtime is simultaneously what makes the filter effective and what pushes the policy out of distribution, so the two cannot be separated by tuning.

Internalisation is attractive because it does separate them. If the corrections shape the policy's behaviour during training, their effect can survive without a correction occurring at deployment — and the resulting policy can still be shielded where a safety case demands a runtime guarantee, so the two mechanisms are complementary rather than exclusive. The question is therefore not only whether the shield can be removed, saving the inference budget it consumes and the obstacle geometry it requires at deployment. It is whether the safety a runtime filter supplies can be had without the interference it imposes.

1.3 Rationale

Research aim. This research aims to determine whether the collision avoidance supplied by an inference-time CBF shield can be transferred into a VLA policy and characterize the mechanism under which this might be achieved.

Research objectives. The following objectives guide the study:

1. To implement a CBF shield over $\pi_{0.5}$ and quantify its effect on collision and task success against the published baseline for the SafeLIBERO benchmark.
2. To distil the shielded policy's own corrected rollouts into its action head, and to measure the resulting policy's safety and capability with the shield removed at inference.
3. To isolate the mechanism responsible for any transfer, through controlled ablations that vary the source of the demonstrations and the presence of the shield during collection.
4. To characterise the limits of the transfer: which collisions persist and why.

Research questions. These objectives translate into four questions, each addressed in a corresponding section of Chapter 4:

RQ1. Does a CBF shield make a pretrained VLA policy safe on this benchmark, and how do the resulting figures compare with published results? (Section 4.3)

RQ2. Can that safety be internalised by distillation, such that the policy remains safe with the shield removed, and at what cost to task capability? (Section 4.4)

RQ3. What determines whether the transfer occurs — what mechanism produces it, and does the source of the demonstrations matter? (Sections 4.5 and 4.6)

RQ4. What does the transfer fail to cover? (Sections 4.7)

1.4 Scope

This study is conducted entirely in simulation, on the `SafeLIBERO` benchmark. No physical hardware was used and no claim of sim-to-real transfer is made.

The policy under study is $\pi_{0.5}$, running in an operational-space action representation with a flow-matching action head. A single policy was chosen to allow comparisons between demonstration sources for the same student.

The barrier is constructed from obstacle geometry read directly from the simulator rather than from perception. Estimated geometry would introduce perception error into the corrections being distilled, and the attribution of failure to transfer would be confounded. Removing perception error makes the result interpretable, at the cost that the shielded figures reported here are an upper bound for this class of filter.

Within that setting, safety carries the narrow meaning given in Section 1.1. The obstacles are static and rigid, so the study does not address moving obstacles, human proximity, contact-force limits, or self-collision, each of which would need a different barrier and, in some cases, a different sensing arrangement.

1.5 Significance

The practical case for this work rests on how learned manipulation policies are actually deployed. A VLA model cannot be certified by inspection, so any deployment near people or expensive equipment places a safety component in front of it — and that component then becomes part of the system that must be justified, maintained, and accounted for when it fails. This study offers three findings that bear directly on that arrangement. Distilling before shielding produces the lowest collision rate of any configuration measured, at a task success rate indistinguishable from shielding the base policy, so where a runtime constraint is required it is better applied to a distilled policy than to a raw one. The distilled policy also requires the shield roughly half as often, which matters wherever motion must be predictable: a filter that fires constantly is a filter that makes cycle times vary. And safe demonstrations are approximately twice as cheap to collect with the shield active than without it, which is a direct cost saving for any deployment gathering task-specific data.

None of this means a VLA policy is ready to run unshielded on a factory floor. What it establishes is that the mechanism works: safety supplied by an external filter can be moved into a policy's weights, in a setting where that filter is the current state of the art, and the gains survive removal of the filter on held-out conditions. That makes the remaining work characterised rather than open-ended: a barrier covering the whole robot body rather than principally the end-effector, geometry estimated from perception rather than read from a simulator, and validation on physical hardware in the target environment.

Academically, the study addresses a question the literature has left open. Runtime filtering and learned safety representations both retain a corrective component at deployment, and the results showing that safety can be transferred into weights either require conditions large visuomotor policies do not satisfy or distil from teachers derived from the student itself. This work joins the two, and in doing so establishes a boundary condition: the transfer depends on where the supervision sits rather than on who produced it.

1.6 Structural Overview

Chapter 2 reviews the literature in three strands: runtime safety filtering for learned policies, work on what makes corrective demonstrations transferable into a policy's weights, and policy-level alternatives that pursue safety through constrained learning instead of correction. It closes by locating the gap this study addresses and stating precisely what is and is not new about it.

Chapter 3 describes the benchmark and metrics, the geometric construction of the barrier and the shield built from it, the two demonstration sources that are compared, and the distillation procedure. Chapter 4 reports the experiments, organised by research question, and includes the controls used to isolate the mechanism.

Chapter 5 interprets those results against the literature of Chapter 2, establishing what each finding contributes and where it sits against prior work. Chapter 6 returns to the research questions and states what the study answers, the limitations that bound those results, the research they recommend, and what the study contributes.

Chapter 2

Related Work

2.1 Introduction

This chapter reviews the research relevant to the aim stated in Section 1.3: determining whether the collision avoidance supplied by an inference-time control barrier function shield can be amortised into the weights of a vision-language-action policy. Its purpose is to establish what is already known about making learned policies physically safe, to identify what remains unresolved, and to position the study against the work it builds on.

The review covers three bodies of work: runtime filtering, in which an external component corrects a policy's actions as they are produced; methods that transfer safety-relevant behaviour into a policy's parameters through imitation; and policy-level approaches that pursue safety through constrained learning rather than filtering.

The chapter is organised by approach. Section 2.2 presents the control-theoretic framework the study instantiates. Sections 2.3.1 to 2.3.3 synthesise the empirical work, moving from methods that keep a corrective component at deployment to those that attempt to remove it. Section 2.4 states the resulting gap and the type of gap it is.

2.2 Theoretical Framework

Two established bodies of theory underpin this study. The first defines what it means for a controller to be safe and supplies the mechanism that enforces it. The second determines whether behaviour enforced by that mechanism can be transferred into a policy by supervised learning.

2.2.1 Safety as Forward Invariance

Section 1.1 introduced the barrier informally; this section gives the form the shield of Chapter 3 instantiates. A control barrier function encodes a safe set $\mathcal{C}$ as the superlevel set of a differentiable function h, and converts safety into a condition affine in the control input: any u satisfying $\dot{h}(x, u) \geq -\alpha(h(x))$ renders $\mathcal{C}$ *forward invariant*, meaning a system that begins inside the set never leaves it [1].

Because that condition is affine in u, it can be imposed as one linear inequality per constraint inside a quadratic program that stays as close as possible to a nominal input:

$$u^\star = \arg\min_u \tfrac{1}{2}\|u - u_{\mathrm{nom}}\|^2 \quad \text{s.t.} \quad \nabla h^\top (f + gu) \geq -\alpha(h). \tag{2.1}$$

This is the object the thesis calls a *shield*: a filter that accepts whatever a nominal controller proposes and returns the minimum modification satisfying the constraint.

Two properties make it usable as a teacher rather than merely as a guard. It is agnostic to how the nominal action was produced, so it composes with a policy it knows nothing about. And because it minimises $\|u - u_{\mathrm{nom}}\|$, its output is not an arbitrary safe action but the *closest* safe action to the one the policy wanted — which is precisely the form a supervised learning target should take.

That second property is why this formulation was adopted over the alternatives. A constrained Markov decision process expresses safety as a bound on expected cost and yields an optimised policy, not a per-step correction, so it produces nothing to imitate. A reward penalty expresses safety as a scalar added to the objective, leaving the policy to infer which of several hundred actions was responsible for a cost it incurred. Only the barrier formulation returns a corrected *action*, at the step where the correction was needed, in the space the policy emits.

2.2.2 Imitation and Covariate Shift

Whether such corrections transfer is a separate question, governed by a different theory. *Behaviour cloning* treats imitation as supervised learning: minimise the discrepancy between the learner's action and the demonstrator's, over a dataset of demonstrated state-action pairs. Its weakness is well characterised. The learner is trained under the demonstrator's state distribution but evaluated under its own, and because control is closed-loop, small errors move it to states the demonstrator never visited, where its error is unconstrained and compounds — a failure mode that grows quadratically rather than linearly in the task horizon [12].

The established remedy is to change where the supervision is collected rather than how much of it there is: query the expert at the states the *learner* visits, and aggregate those labels into the training set [12]. Applied to safety specifically, the same idea motivates querying a safe controller precisely when the learner is about to need it [18].

This is the theoretical basis for a prediction the study tests directly. If what makes corrective supervision transferable is the state distribution it occupies, then two demonstration sets of equal size and equal correction density should differ in effect according to where their states sit. Section 4.6 constructs exactly that comparison.

2.2.3 Two Limits on What May Be Claimed

The guarantee of Equation 2.1 is continuous-time and assumes known control-affine dynamics, so a discrete-time implementation on a contact-rich manipulator satisfies it only approximately (Section 3.5). More fundamentally, forward invariance is a property of the *closed loop containing* the quadratic program, not of the policy inside it. It is therefore not a quantity that can be extracted and stored in weights. This is why the question of Chapter 4 is empirical rather than a corollary, and why its result is reported as a measured reduction in collisions rather than as a guarantee.

2.3 Empirical Research

2.3.1 Filtering at Runtime

The most direct route to a safe learned policy is to place a corrective component in front of it, and the strongest evidence that this works comes from the study that introduced the benchmark used here. AEGIS inserts a safety-constraint layer between $\pi_{0.5}$ and the robot: a vision-language module grounds obstacle geometry from observations, and a CBF-QP modifies the policy's proposed action whenever it violates the resulting constraint [8]. The effect is large — collision avoidance rises from under 19% to between 69% and 78% depending on the action space, while task success improves rather than degrades — and the cost is modest, at 0.356 ms per control step.

Learned variants pursue the same end without an analytic barrier, and their differences are instructive. ConBaT trains a safety critic from binary safe/unsafe labels and operates in embedding space rather than state space, so it applies where explicit geometry is unavailable; at deployment it rectifies actions by backpropagation through that critic until the barrier conditions hold [10]. Latent Policy Barrier instead treats the latent manifold of expert demonstrations as an implicit barrier, training a dynamics model on both expert data and the policy's own rollouts, then steering in latent space at inference [14]. The two differ in what defines the safe set — a learned critic against a demonstration manifold — but converge on the same deployment architecture.

That convergence is the pattern worth extracting. Across all three, safety is supplied by a component that runs at inference and leaves the policy itself unmodified. Latent Policy Barrier makes this explicit, advertising plug-and-play compatibility with pretrained policies without retraining or fine-tuning. The consequence is that the question this thesis asks cannot be posed within this line of work: removing the component returns the base policy by construction, because nothing was ever transferred into it.

2.3.2 Transferring Safety into Weights

A second body of work does modify the policy, and asks under what conditions safety-relevant behaviour survives in its parameters. Cosner et al. establish that it can, formally: imitating a CBF-based expert yields a learned controller with input-to-state-safety guarantees, provided the expert is a *robust* barrier program with explicit disturbance margins, the dynamics are known and control-affine, and the learned controller satisfies conditions on data density along the safe-set boundary, bounded learning error, and a known Lipschitz constant [4]. Even then the guarantee holds on an expanded safe set that degrades with data sparsity. The demonstrations are low-dimensional — an inverted pendulum and a vehicle on a track — and no large visuomotor policy satisfies the stated conditions. What survives for this thesis is not the theorem but its framing: the goal is to transfer safety *guarantees* rather than to reproduce the expert's behaviour, a distinction Section 4.7 observes empirically.

Where formal conditions are unavailable, the empirical literature converges on a different answer: what matters is not the fidelity of the imitation but the location of the supervision. SAFE-GIL injects reachability-computed adversarial disturbances during collection, steering a fixed expert toward the states a learner's errors would produce so that its recovery behaviour is recorded [3]. Its ablation is the decisive evidence: Gaussian noise, uniform noise and DART all fail to reproduce the benefit, so it is the *targeting* of the disturbance rather than augmentation as such that carries the effect. IntervenGen arrives at the same conclusion by a different route, rolling out the current policy so that recorded mistakes are its genuine failures, then transforming human recovery segments onto the states reached, and reporting that ten interventions expanded synthetically outperform a hundred collected directly [7].

In SAFE-GIL the expert is an MPC or PID controller; in IntervenGen the recovery actions come from a human teleoperator. Neither teacher is the learner, and in both cases safety transfers. What the two share is not the identity of the teacher but where its demonstrations sit.

ROAD-VLA distils a proximal teacher — constructed by perturbing the student's own action-token logits with calibrated advantage estimates — along the student's on-policy rollouts, and derives a policy-improvement bound that holds only while the teacher remains close to the current policy [17]. Its privileged-teacher variants fail badly, which is often read as evidence that external teachers cannot transfer. The variants it rejects are, however, all *textual*: a retrieved action hint, an egocentric spatial description, and a task plan. The teacher it accepts is derived from the student itself. No foreign low-level controller appears anywhere in that work, so it neither supports nor refutes the state-coverage account.

One result does bear directly on the protocol this thesis adopts, and it is a negative. Guerrier et al. train a reinforcement-learning agent with a CBF filter active throughout training and then remove it [6]. Safety is not retained: the agent collides once the filter is disabled, and its critic assigns high value *inside* the obstacle, indicating the hazard was never represented at all. Their agent's actions were overridden by the filter, but its learning signal remained a scalar reward, so no gradient ever pointed toward the corrected action. Their curriculum variant, which blends filter and policy actions with the filter's weight decayed to zero, does internalise the constraint. Taken together this establishes that exposure to a shield during training is not by itself sufficient — what the policy is trained *on* decides the outcome.

2.3.3 Constrained Learning as an Alternative

A third line pursues safety through the training objective rather than through filtering or imitation. SafeVLA formulates the problem as a constrained Markov decision process, eliciting unsafe behaviour and optimising the policy under explicit constraints [16]. Safe-Dojo performs model-based safe reinforcement learning inside an action-conditioned video world model, estimating imagined task progress and safety cost separately and optimising them with Lagrangian constrained GRPO; it evaluates on SafeLIBERO, making it the most direct methodological alternative in this literature [15].

These matter as calibration rather than contrast. Neither is scalar-reward reinforcement learning, and both supply substantially richer constraint information than a penalty term.

2.3.4 Methodological Patterns and Open Tensions

Two features of how this literature is evaluated bear on how its findings can be combined. The first is that the reported quantities are not mutually comparable. AEGIS defines collision avoidance as the proportion of strictly collision-free episodes and states no numeric tolerance [8]; SAFE-GIL reports collision rates on two-dimensional navigation and aircraft taxiing [3]; ROAD-VLA reports task success under distribution shift [17]. Sample sizes range from tens of trials to sixteen hundred episodes, and single-seed or few-seed reporting is common. Cross-paper numerical comparison is therefore mostly unavailable, which is why this study reproduces its baseline directly rather than quoting a published figure (Section 4.3).

The second is scale. The transfer results that are best characterised are also the furthest from the setting studied here: multilayer perceptrons on low-dimensional state in SAFE-GIL [3], a unicycle model in a 1.5 m arena in Guerrier [6], an inverted pendulum in Cosner [4]. Whether their conclusions hold for a three-billion-parameter flow-matching policy is an open empirical question rather than a settled one.

Against that, the field agrees on three things. Runtime filtering substantially reduces collisions for learned manipulation policies. Corrective demonstrations transfer more effectively than additional clean ones. And scalar rewards are a poor channel for safety, a conclusion reached independently by studies of CBF-guided reinforcement learning [6] and by the constrained formulations that avoid them [15, 16].

One tension remains unresolved, and it is the one this study addresses. SAFE-GIL and IntervenGen show foreign teachers transferring safety; ROAD-VLA is widely read as showing they cannot. The readings are reconcilable only if the operative variable is where the supervision sits rather than who produced it — and no existing study varies the teacher's state distribution while holding the correction signal fixed. A second, smaller tension concerns combination: SAFE-GIL reports filtering and corrective imitation as complementary without much degradation in task performance, a claim Section 4.4 is positioned to test directly.

2.4 Research Gap

The gap this study addresses is primarily methodological, with a secondary theoretical dimension. It is methodological because the existing work relies on a single architectural arrangement — safety supplied by a component that runs alongside the policy — and the alternative arrangement, in which the corrections that component produces are compiled into the policy itself, has not been evaluated. It is theoretical because the framework that would predict when such a transfer succeeds has been developed and tested only at a scale far removed from contemporary vision-language-action models.

The methodological gap is structural rather than an oversight, and this distinction matters. In the filtering literature the policy is never modified: AEGIS corrects actions as they are produced [8], ConBaT rectifies them by backpropagation at test time [10], and Latent Policy Barrier steers in latent space while explicitly requiring no retraining [14]. Removing the component from any of these returns the base policy unchanged, so there is no transfer to measure. The question cannot be posed within that line of work.

Where a policy *is* shaped under a filter, the one available result is negative. Guerrier et al. train under an active barrier and then remove it, and find the constraint was never internalised [6]. That study is the closest existing analogue to the present one, and it differs in a single respect: its learning signal was a scalar reward, so the corrected action was imposed on the policy but never presented to it as a target. Whether the same protocol succeeds when the correction is the imitation target has not been tested. The gap is therefore not that nobody looked, but that the one attempt used a channel the corrections could not travel through.

The theoretical dimension follows from Section 2.2. Conditions under which a learned controller inherits safety from a barrier-based expert are established [4], but demonstrated on an inverted pendulum and a vehicle on a track, under assumptions — known control-affine dynamics, bounded learning error, a known Lipschitz constant — that a three-billion-parameter flow-matching policy does not satisfy. Separately, covariate-shift theory predicts that supervision transfers according to the states it occupies [12], a prediction the empirical literature supports indirectly [3, 7] but has never tested by varying the state distribution while holding the correction signal fixed.

This study addresses that gap by constructing the missing arrangement and the missing comparison: distilling a control-theoretic shield's own corrections into the policy that generated them, evaluating the result with the shield removed, and contrasting it against a teacher matched in correction density but not in state distribution. The claim of novelty is correspondingly narrow. Learning safety from barrier-based supervision is not new [4], and distilling a corrective signal into a vision-language-action policy is not new [17]. What has not been done is the conjunction — a control-theoretic filter, distilled from a policy's own shielded rollouts, evaluated with the filter removed, in the setting where runtime shielding is the current state of the art — together with the boundary condition that establishes what the transfer depends on. This study does not resolve the question of how to make vision-language-action policies safe; it establishes that one mechanism transfers, and identifies what governs whether it does.

2.5 Conclusion

This chapter has reviewed three bodies of work bearing on the safety of learned manipulation policies. Runtime filtering, whether through an analytic barrier or a learned one, reliably reduces collisions but supplies safety through a component that runs alongside the policy and leaves it unchanged. Work on transferring safety into a policy's parameters establishes that corrective demonstrations can carry it, and converges on the location of the supervision rather than the identity of the teacher as the operative variable — though the results that characterise this best are drawn from settings far smaller than a contemporary vision-language-action model. Constrained learning offers a third route, pursuing safety through the training objective rather than through correction at all. Underpinning the first of these is the control-theoretic account of safety as forward invariance; underpinning the second is the theory of covariate shift in imitation learning.

The gap that follows is methodological, with a theoretical dimension. No study has taken the corrections a control-theoretic filter produces, compiled them into the policy that generated them, and evaluated what remains once the filter is removed — and the one study that trained a policy under an active filter used a learning signal through which those corrections could not travel.

Addressing that gap requires three things of the method: a shield whose corrections are substantial enough to constitute a training signal, two demonstration sources matched in correction density but differing in the states they occupy, and a distillation procedure that can be evaluated with the filter both present and absent. Chapter 3 sets out each in turn.

Chapter 3

Method

3.1 Introduction

The aim as stated in Section 1.3 is to determine whether the collision avoidance supplied by an inference-time CBF shield can be transferred into a VLA policy and characterize the mechanism under which this might be achieved. Section 2.5 identified three things a method must supply to test that. This chapter addresses each, and the reasoning behind the design decisions they involve.

The chapter proceeds from the general to the specific. Section 3.2 states the research design and the variables it holds fixed. Section 3.3 describes the benchmark, the metrics, and the division of initial states into training and evaluation. Sections 3.4 and 3.5 construct the barrier and the quadratic program built from it, and state the approximations each involves. Section 3.6 describes the two demonstration sources whose comparison is the central experiment, and Section 3.7 the procedure that trains on them. Section 3.8 defines how the resulting measurements are compared.

3.2 Research Design

The study uses a controlled experimental design. The research questions are causal and only a design that isolates variables can answer them.

Four comparisons make up the study, each manipulating a single variable. The first varies whether the shield runs at inference, establishing that its corrections are large enough to constitute a training signal. The second varies whether the policy has been distilled, and measures the result with the shield removed. The third varies the source of the demonstrations while holding their number, filtering criteria and correction density fixed. The fourth varies whether the shield was active during collection, holding the demonstration count and the retention criteria constant.

Across all four, the following are held fixed: the base checkpoint, the training objective and optimiser configuration, the number of gradient steps (matched to within one percent, Section 3.7), the evaluation code, the scenes, and the held-out initial states on which every condition is evaluated. Action sampling is deterministic except where a comparison requires otherwise. This degree of control is only achievable via simulation.

3.3 The SafeLIBERO Benchmark

All experiments use SafeLIBERO [8], which augments the LIBERO manipulation suite [9] with obstacles placed in the workspace. The grid is three task suites (spatial, object, goal) $\times$ two obstacle levels $\times$ four tasks, giving 24 distinct scenes, each providing 50 randomised initial states.

Experiments run in MuJoCo through the robosuite interface that LIBERO builds on [9]. The platform is a 7 DoF Franka Emika Panda with a parallel-jaw gripper, controlled in operational space. Observations are a 224×224 agent-view image, a wrist camera at the same resolution, and eight proprioceptive dimensions: end-effector position, orientation as an axis-angle, and the two gripper finger positions. Notably the observation contains no joint angles, a fact Section 4.5 returns to.

Initial states 0–34 are used for demonstration collection and training; states 35–49 are held out and used only for evaluation. The evaluation varies object placement within trained scenes rather than introducing new tasks or objects, so it tests generalisation of avoidance across configurations, not across tasks.

Every reported condition uses five held-out initial states from each of the 24 scenes, giving $n = 120$ rollouts and each rollout is limited to a maximum of 300 control steps.

3.3.1 Metrics

Task success rate (TSR) is the fraction of rollouts in which the environment's own goal predicate is satisfied at any point before the horizon expires. Success is binary per rollout.

Collision is recorded when the obstacle's position departs from its initial position by more than one millimetre in the ℓ_1 sense,

$$\sum_{k \in \{x,y,z\}} \big| p_k(t) - p_k(0) \big| > 0.001 \text{ m}, \tag{3.1}$$

and a rollout is marked collided if this holds at any control step. Collision-avoidance rate (CAR) is the complement.

The reference pose $p(0)$ is taken after a brief settling period, since objects spawned in unsupported poses would otherwise cross the threshold under gravity before the policy has acted.

Execution time to success (ETS) is the control step at which the success predicate first holds, averaged over successful rollouts only and reported in control steps. ETS is therefore conditional on success and must be read alongside TSR.

CBF activation rate is the fraction of control steps on which the shield altered the action, counted whenever the projected action differs from the nominal by more than 10^{-4} in norm.

Mean correction norm is the average $\|u^\star - u_{\mathrm{nom}}\|$ over steps where the shield intervened, measured against actions bounded to $[-1, 1]$ per axis. This metric reports the magnitude of the corrections.

Per-body attribution records which of the robot's bodies contacted the obstacle in each episode, grouped as gripper, arm link, held object, and other scene objects.

3.4 Obstacle Geometry and the Barrier

The obstacle's collision mesh is read from the simulator, sampled into a point cloud, and decomposed into a small set of spheres.

End-effector. Three spheres in the end-effector frame approximate the Franka hand: a palm sphere of radius 4.8 cm at 5.6 cm behind the grasp site, and two finger-pad spheres of radius 2.0 cm offset 3.6 cm either side at 10.5 cm depth.

Arm links. Links 3-7 and the palm body are each assigned a radius (5.5 cm to 9.0 cm, set from the link's maximum transverse extent plus a 15–21 mm margin) and sampled at three positions along the link axis, so a long link is not represented by its origin alone.

For each pair of a robot sphere i and an obstacle sphere j the barrier is

$$h_{ij} = \|p_i - c_j\|^2 - (r_i + r_j)^2, \tag{3.2}$$

with $h > 0$ the safe set. Differentiating along the rigid-body velocity gives the constraint row

$$\left[\, 2\, s\, (p_i - c_j)^\top R_1 \,\right] u \;+\; k\, h_{ij} \;\geq\; 0, \qquad k = 10, \tag{3.3}$$

where u is the commanded end-effector velocity in the body frame, R_1 its rotation and s a scale factor. For arm links, whose velocity is not the commanded one, the mapping $v_{\mathrm{link}} \approx J_{\mathrm{link}} J_{\mathrm{ee}}^{+}\,(s R_1 u)$ is used, with J_{ee}^{+} a damped pseudo-inverse ($\lambda = 10^{-3}$).

3.5 The CBF Shield

At each control step the policy's proposed translation is treated as a nominal end-effector velocity u_{nom}, and the shield solves

$$u^\star = \arg\min_u \|u - u_{\mathrm{nom}}\|^2 \quad \text{s.t.} \quad a_m^\top u + b_m \geq 0 \;\text{ for every constraint row } m, \tag{3.4}$$

a quadratic program in three variables whose rows are the end-effector/obstacle and arm-link pairs of Section 3.4. This is the standard CBF-QP form [1], instantiated on the sphere-pair barrier of Equation 3.2. When no constraint is active the solution is u_{nom} and the action passes through unchanged.

The problem is solved with OSQP [13] through CVXPY [5]. An intervention is recorded when the returned action differs from the nominal by more than 10^{-4}.

Two discrete-time adjustments are required because the barrier's guarantee is continuous-time. The effective radius is inflated by a lag buffer covering the distance travelled between control steps and the inverse-kinematics tracking error. Rotation is not constrained, the barrier acts on the translational channel only.

3.6 Teachers

Two sources of demonstrations are compared, and the comparison between them is the central experiment of this work.

3.6.1 The Shielded Policy as Its Own Teacher

The policy is rolled out with the shield active, and episodes are retained if they succeeded, displaced nothing, and the shield intervened at least once.

3.6.2 A Privileged Scripted Expert

The alternative teacher is a sampling-based motion planner: RRT-Connect in joint space with clearance inflation, followed by inverse kinematics and playback as operational-space deltas. It reads object and goal poses directly from the BDDL scene specification [9].

The planner produces collision-free plans, but not collision-free executions: inverse-kinematics tracking error and operational-space control cause the realised trajectory to depart from the planned one. Its demonstrations are therefore collected under the same shield as the policy's own (Section 3.5). The two teachers differ in the states their demonstrations occupy, not in whether they were corrected — the planner plans from its own initial conditions and is never conditioned on where $\pi_{0.5}$ fails.

3.7 Distillation

Retained demonstrations are behaviour-cloned using the model's native flow-matching loss [2, 11]. The learning process uses a LoRA adapter on the action head and freezes the vision-language backbone. The imitation targets are the corrected action sequences. Actions are normalised by passing them through the policy's own input transform, so the target matches the distribution the model was pretrained on rather than a hand-rolled normalisation. Because the corrections are applied to actions the policy itself proposed, they are located by construction at states the policy reaches — the property Section 2.2 identifies as governing whether supervision transfers.

The procedure sits in the DAgger lineage [12], and shares with SafeDAgger [18] the idea that a safe controller's interventions are themselves the informative signal.

Training Control and Alignment When comparing distilled policies, training runs are matched strictly on the total number of gradient steps rather than the number of epochs, controlling for disparities in episode lengths across demonstration sources. The distilled policies compared are matched at 16,140 and 15,988 gradient steps respectively.

3.8 Analysis

Unless stated otherwise, every headline quantity is a proportion over $n = 120$ independent rollouts, reported with a 95% Wilson score interval. Comparisons are made by interval overlap: a difference is reported as established when the two intervals are disjoint, and as suggestive when they overlap.

Results are pooled across the 24 scenes. Pooling is appropriate because the manipulated variable is the same in every scene and the interest is in its average effect. Where the per-scene distribution carries information that the pooled figure hides, it is shown separately, as in the coverage comparison of Section 4.6.

Two secondary analyses support the main comparisons. Per-body collision attribution (Section 3.3.1) decomposes the residual by which part of the robot made contact. And demonstration sets are characterised by activation rate and correction norm.

3.9 Conclusion

The design is a controlled experimental comparison in which a single variable is manipulated at a time and the apparatus is otherwise held fixed. That apparatus comprises a benchmark of 24 scenes with a training and evaluation split over initial states, a sphere-decomposition barrier over the end-effector and arm links, a quadratic program that projects the policy's proposed action onto the resulting safe set, two demonstration sources differing in the states they occupy, and a behaviour-cloning procedure matched across conditions on gradient steps. Measurements are proportions over held-out rollouts, compared by interval overlap.

Each element answers to one of the three requirements set out at the start of the chapter. The shield supplies corrections dense and large enough to constitute a training signal; the two teachers differ in state distribution while matching in correction density; and the distillation target is the corrected action itself, so the policy can be evaluated with the filter both present and absent. Chapter 4 reports what that yields.

Chapter 4

Experiments and Results

4.1 Introduction

A VLA policy that is competent at manipulation is not thereby safe, and safety filters are a mitigation not a solution. This study asks whether those corrections can instead be compiled into the policy. (Section 1.3).

The method is the controlled comparison set out in Chapter 3: a shield is built over $\pi_{0.5}$, its corrections are collected as demonstrations, those demonstrations are behaviour-cloned back into the policy's action head, and the result is evaluated with the shield removed.

This chapter presents what those experiments produced. It reports the measurements and states what each contributes to the research questions of Section 1.3.

The chapter follows the research questions in order. Section 4.3 establishes whether the shield works at all (RQ1) and Section 4.4 whether its effect survives distillation (RQ2). Sections 4.5 and 4.6 address what determines whether the transfer occurs, first by isolating the shield from outcome selection and then by varying the demonstration source (RQ3). Section 4.7 reports what the transfer does not cover (RQ4).

4.2 The Demonstration Sets

Three demonstration sets were collected, and every distillation result reported in this chapter draws on one of them. Their composition is given in Table 4.1.

Table 4.1: The three demonstration sets.

Set	Demos	Rollouts	Yield	Scenes	Activation	Corr. norm
Shielded self-rollouts[a]	186	480	39%	24	0.32	0.33
Privileged planner[b]	318	864	37%	18	—	—
Unshielded control	85	480	18%	24	0	—

[a] Activation is the median fraction of control steps shielded.
[b] Six of the 24 scenes produced no usable demonstrations (Section 4.6).

4.3 Does the Shield Make $\pi_{0.5}$ Safe?

The first question is whether a CBF shield, applied as a runtime filter, makes a pretrained VLA safe on this benchmark at all. Table 4.2 reports the comparison: the shield removes 69 percentage points of collision, with disjoint confidence intervals, while raising success from 58.3% to 71.7%.

Table 4.2: Safety and task performance for base $\pi_{0.5}$ with and without shield.

Policy	**TSR (95% CI)**	**Collision rate (95% CI)**	**CAR**
Base, no shield	58.3% [49.4, 66.8]	82.5% [74.7, 88.3]	17.5%
Base + shield	71.7% [63.0, 79.0]	13.3% [8.4, 20.6]	86.7%

Answering RQ1. A CBF shield does make a pretrained VLA policy safer on this benchmark: collisions fall by 69 percentage points with disjoint intervals, and task success is not only maintained but increased. The corrections it applies are therefore large enough to constitute a training signal, which is the precondition for everything that follows.

4.4 Can the Shield's Safety Be Internalised?

The shielded demonstration set of Table 4.1 was behaviour-cloned into the policy's action head, and the procedure was repeated for a second round.

Table 4.3: Effect of self-distillation, with and without the shield at inference. Rows in bold are the shield-free distilled policies.

Policy	**TSR (95% CI)**	**Collision (95% CI)**	**CAR**	**CBF act.**[a]	**ETS**
Base, no shield	58.3% [49.4, 66.8]	82.5% [74.7, 88.3]	17.5%	—	138.6
Base + shield	71.7% [63.0, 79.0]	13.3% [8.4, 20.6]	86.7%	0.355	160.6
Round 1, no shield	**82.5% [74.7, 88.3]**	**19.2% [13.1, 27.1]**	**80.8%**	—	157.3
Round 1 + shield	70.8% [62.2, 78.2]	8.3% [4.6, 14.7]	91.7%	0.177	170.0
Round 2, no shield	**80.8% [72.9, 86.9]**	**17.5% [11.7, 25.3]**	**82.5%**	—	158.8
Round 2 + shield	72.5% [63.9, 79.7]	9.2% [5.2, 15.7]	90.8%	0.182	176.7

[a] CBF activation rate.

With no shield running at inference, the distilled policy displaces the obstacle in 19.2% of episodes, against 82.5% for the policy it was distilled from — and its success rate rises from 58.3% to 82.5%. The distilled policy is simultaneously safer and more capable than its own teacher, without the teacher's filter. Most of the shield's safety benefit survives its removal.

Rounds one and two overlap on both metrics, so a single round is enough. It also does not erode performance. Applying the shield to the distilled policy lowers collisions further, to 8.3%, but reduces success from 82.5% to 70.8%. ETS rises from 138.6 to 157.3 steps, roughly 14% slower. The distilled policy takes a more conservative route.

Answering RQ2. The shield's safety is internalised. With no shield at inference the distilled policy collides in 19.2% of episodes against the base policy's 82.5%, and succeeds in 82.5% against 58.3%, both intervals disjoint. It recovers most of the shielded baseline's collision avoidance (80.8% against 86.7%) without the filter, in a single round, at a cost of roughly 14% in execution time.

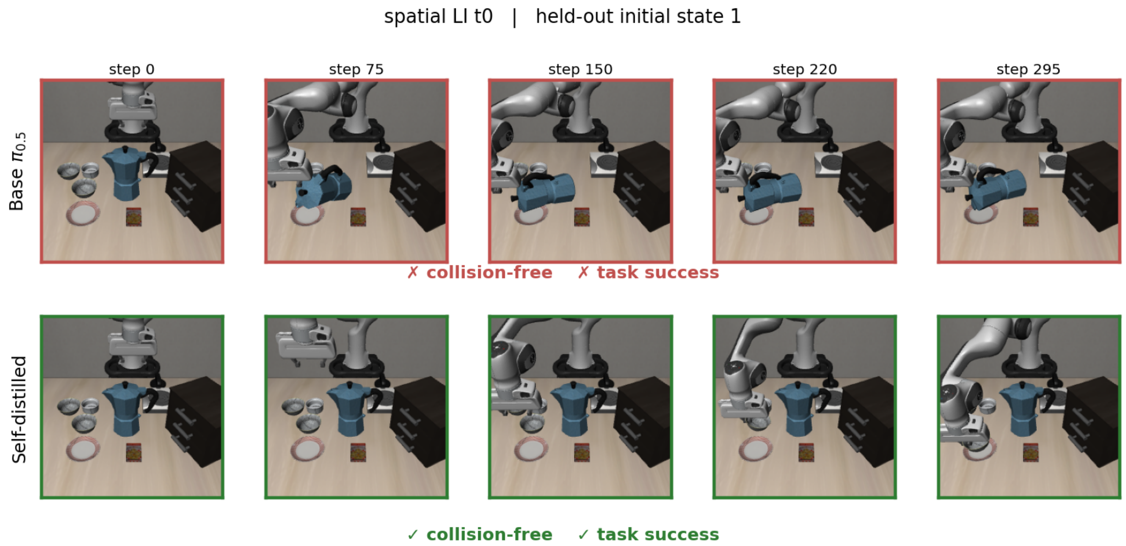

Figure 4.1: The same scene under the base and distilled policies

4.5 What Explains the Improvement?

Section 4.4 showed that distilling the shielded policy's own clean rollouts makes it dramatically safer without the shield; however, it does not explain why this is so. The demonstrations were filtered on three properties at once — the shield fired, the episode succeeded, and nothing was displaced — and two mechanisms are conflated in the filter.

Under the first, distillation, the policy imitates the shield's avoidance corrections. This is the claim of this work. Under the second, selection, the improvement comes from training a policy on its own successful episodes, a known self-improvement effect requiring no shield. If the second mechanism accounts for the result, the shield is incidental and the contribution is a rediscovery of filtered behaviour cloning.

To disambiguate this, a matched control was run.

Table 4.4: The matched control. Both distilled conditions use 85 demonstrations.

Condition	TSR (95% CI)	Collision (95% CI)
Undistilled base	58.3% [49.4, 66.8]	82.5% [74.7, 88.3]
Control, shield **off**	50.0% [41.2, 58.8]	84.2% [76.6, 89.6]
Matched, shield **on**	**75.8% [67.4, 82.6]**	**26.7% [19.6, 35.2]**

Compared to the base policy, the control has a marginally worse collision rate and more substantial TSR degradation. So, success-filtering by itself is unsuccessful in this benchmark. At the same demonstration count and under the same filter, the shielded condition reaches 26.7%, with a confidence interval disjoint from the control's. Therefore, the difference between the two conditions is attributable to the shield. The activation rate corroborates this independently: after distillation the shield finds materially less to correct (Table 4.3), and a policy that merely succeeded more often would not require *less* safety intervention.

The control is not uniformly inert, however. Per-body attribution shows its arm-link collisions falling from 17 to 6 while gripper collisions remain at 44, which is why the total did not move. Outcome selection therefore transfers avoidance in the channel the barrier does not express, while leaving the channel that dominates the collision count unchanged.

Answering RQ3, first part. The improvement is produced by the shield's corrections, not by training on successful episodes. At matched demonstration count and under an identical filter, collection with the shield switched off leaves the policy indistinguishable from the undistilled baseline, while collection with it active produces a substantial improvement.

4.6 Does the Source of the Demonstrations Matter?

Section 4.5 established that the shield's corrections, rather than outcome selection, produce the transfer. This section asks whether it matters whose rollouts those corrections are applied to. A sampling-based motion planner was used as an alternative teacher: RRT-Connect in joint space with clearance inflation, followed by inverse kinematics and playback as operational-space deltas (Section 3.6.2).

Table 4.5: Comparison of demonstration sources for policy distillation.

Policy	**TSR (95% CI)**	**Collision rate (95% CI)**
Base, no shield	58.3% [49.4, 66.8]	82.5% [74.7, 88.3]
Planner-distilled, no shield	54.2% [45.3, 62.8]	80.0% [72.0, 86.2]
Self-distilled (round 1), no shield	**82.5% [74.7, 88.3]**	**19.2% [13.1, 27.1]**

The planner-distilled policy is statistically indistinguishable from the undistilled base policy on both metrics.

Correction density in the two demonstration sets. Both sets were collected under the same shield (Section 3.6.2), so the planner's demonstrations are corrected as the policy's own are. That the planner depends on those corrections is measurable: run unshielded on spatial task 0 it collides in 5 of 8 episodes, with contact from arm links, gripper, held object and scene objects alike, and in none when shielded.

State coverage. Figure 4.2 takes the base policy's own visited states and asks how closely each demonstration set approaches them.

Subsampled to equal size per scene, the shielded rollouts cover 37.7% of the policy's own states within 2 cm against the planner's 22.8%, and cover better in 15 of the 18 scenes holding both demonstration types. Without that control the two sets are near-indistinguishable at 37.7% against 32.8%, so the matching is load-bearing rather than cosmetic.

Two limits apply. Coverage is measured on end-effector position, 3 of the 8 observed dimensions, so it says nothing about joint configuration. And it is descriptive rather than causal: the matched comparison establishes that the demonstration source mattered, while the coverage figure proposes how the two sources differ.

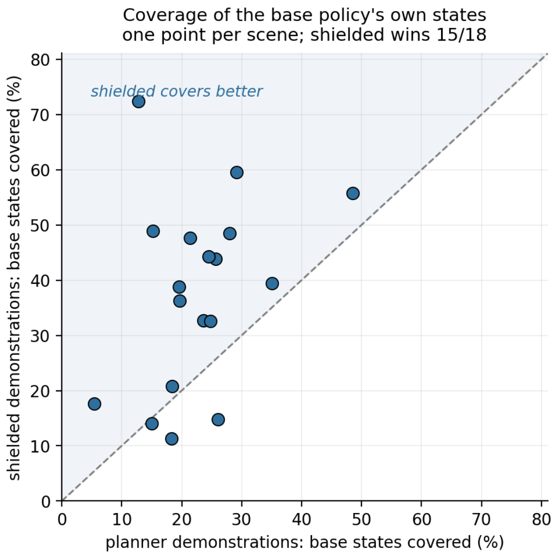

Figure 4.2: Coverage is the fraction of base-policy states lying within 2 cm of a demonstration state. Both demonstration sets are subsampled to equal size per scene, over 30 resamples. One point per scene: the shielded rollouts cover the policy's own states better in 15 of the 18 scenes holding both demonstration types, 37.7% against 22.8% pooled.

Answering RQ3, second part. The source of the demonstrations determines whether the transfer occurs. Corrections of equal density applied to a different controller's trajectories produced no measurable improvement, under a comparison in which the demonstration source was the only variable. The measured difference between the two sources is state coverage; observation aliasing was tested as an alternative explanation and rejected.

4.7 What Can the Shield Not Reach?

The shield significantly reduces collisions but does not completely eliminate them. This section identifies what the residual consists of, using per-body collision decomposition.

In agreement to Section 3.4, the discrete-time approximation holds in practice, eliminating end-effector collisions. Arm-link collisions fall from 17 to 8 between the base and twice-distilled policies, while the shielded baseline still records 15.

Table 4.6: Per-body attribution of collisions in 120 episodes.

Policy	Collided	Gripper	Arm link	Held object
Base, no shield	99	71	17	36
Base + shield	16	**0**	15	0
Round 1, no shield	23	7	13	9
Round 1 + shield	10	**0**	8	1
Round 2, no shield	21	8	8	7
Round 2 + shield	11	**0**	9	0

The likely route is the retention criterion rather than the correction signal. Demonstrations were kept only if nothing at all was displaced, arm links included, so the dataset encodes whole-body avoidance that no constraint row ever expressed. Safety reaches the policy through two channels — imitation of corrections where the filter has authority, and outcome selection where it does not — and only the second can carry behaviour the barrier cannot represent.

Answering RQ4, first part. The transfer does not cover the arm links. Where the barrier's model of the robot is precise and the constrained velocity is the one commanded, collisions are eliminated outright; where the model is coarse and the velocity inferred, they persist, accounting for 15 of the 16 residual collisions under the shield. The distilled policy nonetheless improves on its teacher in that channel, on overlapping intervals.

4.8 Conclusion

This chapter reported the experiments ordered by the research questions. A CBF shield makes $\pi_{0.5}$ substantially safer on this benchmark, and reproduces the published baseline closely enough to validate the evaluation pipeline (RQ1). Distilling the shielded policy's own corrected rollouts internalises that safety: with no shield at inference the policy is both safer and more successful than the model it came from, recovering most of the shielded baseline's collision avoidance in a single round (RQ2).

Two controls bound that result. The improvement is attributable to the shield's corrections rather than to outcome selection, and it does not survive a change of demonstration source: a geometry-privileged planner supplying corrections of equal density produced no measurable gain (RQ3). The transfer leaves the arm links largely uncovered (RQ4).

The following chapter interprets these findings, sets them against the literature of Chapter 2, and states what they do and do not establish.

Chapter 5

Discussion

5.1 Introduction

A VLA policy that is competent at manipulation is not thereby safe, and an external safety filter is a mitigation not a solution. This work asked whether the corrections provided by the filter can instead be compiled into the policy: whether the shield makes $\pi_{0.5}$ safe at all (RQ1), whether that safety survives the shield's removal (RQ2), what determines whether the transfer occurs (RQ3), and what it fails to cover (RQ4). The approach was a controlled comparison on SafeLIBERO, and the results are reported in Chapter 4. This chapter, organized by research question, interprets those results and sets them against the literature of Chapter 2.

5.2 Overview of Key Findings

The data support the claim that a barrier's corrections can be transferred into a VLA policy's weights. Distilling $\pi_{0.5}$ on its own shielded rollouts made it substantially safer and more successful, with no shield at inference, than the base policy. A key pattern to emerge was that safety and competence did not trade off against each other: across every arm evaluated, a lower collision rate occurred with equal or higher task success, the sole exception being the two configurations where the shield was applied to an already-distilled policy.

The analysis also identifies a boundary, and it is the most transferable result. The improvement is attributable to the shield's corrections rather than to training on successful episodes. Corrections of equal density supplied by a geometry-privileged motion planner expert barely transferred at all.

5.3 What Removing the Shield Shows

The shield result itself is a replication: the unshielded baseline reproduces the figures AEGIS reports [8], which works as a sanity-check for downstream experiments.

Guerrier et al. train an agent under an active CBF filter, remove it, and find no internalisation [6]. Exposure to a shield during training is thus demonstrably insufficient on its own.

One core finding is that optimizing for safety also made the policy perform tasks with higher success rate. One reading is that collisions in this benchmark are not purely a safety cost. A displaced obstacle changes the scene the policy is conditioned on, and an episode that has knocked its target aside is frequently no longer completable, so avoiding collisions and completing tasks are partly the same objective. The obvious alternative reading — that the gain is ordinary self-improvement from imitating successful rollouts, with the shield incidental — is excluded by the matched control of Section 4.5, in which distillation on an identically filtered but unshielded set left success *lower* than the undistilled baseline. Filtering for success did not by itself teach success on this benchmark.

5.4 Tradeoffs

The main cost is in execution time. The distilled policy is roughly 14% slower. It takes a more conservative route, and given the focus of this work in safety instead of performance, this is to be expected.

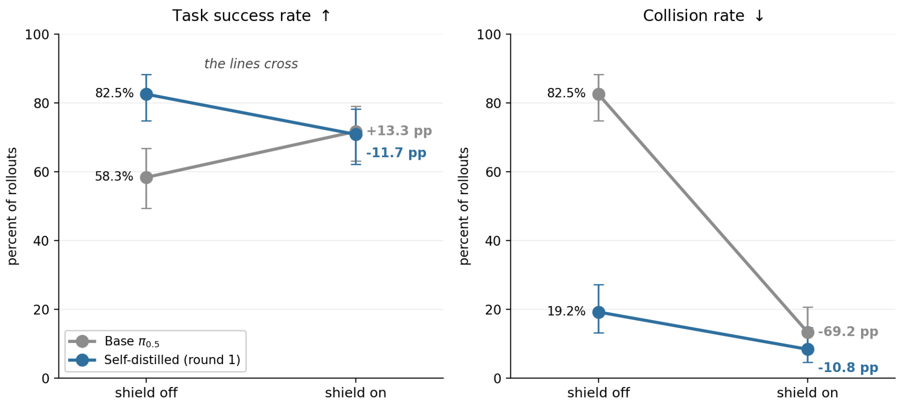

Figure 5.1: Attaching the shield raises the base policy's success and lowers the distilled policy's, but collisions fall in both cases.

There is a cost trade-off that comes from combining mechanisms rather than of distillation itself. As shown in Figure 5.1, applying the shield to the distilled policy improves safety further but reduces success by similar amounts. When the distilled policy already avoids the obstacle, the shield's safe action projection perturbs actions that were already both safe and competent, leading to safety-induced distribution shift. This result is congruent to what is reported in VLSA [8]. That an approach built on runtime enforcement reports the ceiling of runtime enforcement is the strongest available argument for internalisation, and it reaffirms the relevance of self-distillation as an alternative route to safety that retains competence.

The trade runs both ways. Re-attaching the shield costs success but restores the online constraint satisfaction a formal safety argument requires, and it produces the highest collision-avoidance rate of any configuration tested — 91.7%, at a success rate statistically indistinguishable from shielding the base policy. Where a safety case demands a runtime constraint, distilling first is therefore the better starting point. That the residual is 8.3% rather than zero is a reminder that the guarantee is continuous-time and the implementation approximates it (Section 3.4).

5.5 Conditions for safety transfer

The planner teacher experiments showed that dense per-step supervision is necessary but not sufficient. What separated the two teachers were the states they visited: the planner covers the states the student reaches substantially less well (Figure 4.2).

The operative variable is therefore the state distribution the supervision covers, not the identity of the teacher, which is supported by the literature. SAFE-GIL clones an MPC or PID expert and IntervenGen replays human corrections; both teachers are foreign to the learner, and both transfer safety, because both take care that the demonstrations sit at states the learner's errors would produce [3, 7]. Self-distillation is one route to that coverage rather than the only one. Its practical advantage is that it needs no disturbance model or human operator: the policy visits its own failure states unprompted.

5.6 What the Transfer Does Not Reach

The transfer is uneven across the robot, and the pattern is insightful. Collisions were eliminated on the end-effector, and persisted on he arm links, which account for almost all residual collisions under the shield (Section 4.7). What distils is not avoidance in general but avoidance in the channel the barrier expresses well. This is consistent with the barrier being the source of the signal rather than a general safety prior, and it identifies the whole-body barrier as the specific change most likely to raise the ceiling.

5.7 Closing Summary

This chapter interpreted the findings in relation to the four research questions of Chapter 1. The results support the view that a CBF's corrections can be internalised into a VLA policy, leaving it safer and more capable with the shield detached. Safety and competence moved together throughout, except where the distilled policy and the shield were stacked.

The study also identified a clear boundary. The teacher's coverage of the states the student visits is the relevant variable. The transfer extends only as far as the barrier's model of the robot is faithful. These findings carry implications for how safety filters are deployed and for what remains unresolved, which the following chapter draws together.

Chapter 6

Conclusion

6.1 Introduction

This chapter begins with a summary of the key findings in relation to the research questions set out in Chapter 1, followed by the limitations that bound them and the recommendations for future research those limitations imply. It then states the contribution of the study, before closing with a brief overall recap.

6.2 Summary of Findings

This study aimed to determine whether the collision avoidance supplied by an inference-time control barrier function shield can be amortised into the weights of a vision-language-action policy, such that the shield is no longer required at deployment. Under the conditions tested, it can. In relation to RQ1, the shield made $\pi_{0.5}$ substantially safer on this benchmark and reproduced the published baseline closely enough to establish that the evaluation measures what it claims to. In relation to RQ2, distilling the shielded policy's own corrected rollouts reduced collisions from 82.5% to 19.2% **with no shield at inference**, while raising task success from 58.3% to 82.5% on held-out initial states — recovering most of the shielded baseline's collision avoidance in a single round, at a cost of roughly 14% in execution time.

In relation to RQ3, the transfer is bounded, and the bound is the more useful half of the result. The improvement is produced by the shield's corrections rather than by training on successful episodes, but it does not survive a change of demonstration source: a geometry-privileged motion planner, supplying corrections from the same shield, produced no measurable gain under a comparison in which the demonstration source was the only variable. What separates the teacher that worked from the one that did not is where its demonstrations sit. Transferable supervision must correct the learner at states the learner itself reaches.

In relation to RQ4, the transfer does not reach the whole robot. Collisions were eliminated where the barrier's model of the robot is precise and the constrained velocity is the one commanded, and persisted at the arm links, which account for almost all of what remains under the shield. Four alternative mechanisms — scalar-reward reinforcement learning, a language-expressed constraint, best-of-N selection and a generative uncertainty monitor — were each tested and none substituted for shielded imitation, though every one of those negatives is scoped to the implementation tested rather than to its method class.

6.3 Limitations

Simulation only. Every result is from SafeLIBERO. No physical robot was used and no claim of sim-to-real transfer is made.

No formal guarantee is inherited. The distilled policy carries no runtime constraint, so nothing of the shield's formal argument survives its removal. What is demonstrated is an empirical reduction in collisions.

Privileged geometry. The barrier is constructed from obstacle poses and meshes read directly from the simulator, so every shielded figure reported here is an upper bound for this class of filter. The distilled policy uses no geometry at inference, but the dependency remains during training.

Approximate robot model. The barrier represents the robot with spheres rather than its exact shape, because the exact version would not solve fast enough to run at every control step (Section 3.4). How accurate that approximation actually was is never measured directly.

Single training seed. Each distilled policy is one training run and seed variance is unquantified.

Single policy and action space. Everything here uses one model, $\pi_{0.5}$, which produces chunks of actions by integrating a flow field.

Attribution caveats. The per-body attribution is a documented lower bound that can miss delayed or indirect contacts.

6.4 Recommendations for Future Research

A physical robot is the most direct extension. Whether internalised avoidance survives contact dynamics, calibration error and a real perception pipeline is the question this work most needs answered, and Section 5.4 suggests the deployment configuration to test: the distilled policy with the shield reattached, which reached the highest collision-avoidance rate of any arm evaluated while keeping the runtime constraint a safety case would require.

Repeating the training across seeds is the cheapest outstanding item. Each distilled policy here is a single run.

Perception-grounded barriers would remove the privileged-geometry dependency and measure how much of the reported safety survives estimation error — the same question a physical deployment asks, isolated from the rest of the reality gap. Learned safety representations offer one route [10, 14], though both retain the inference-time correction this work removes.

A whole-body barrier with proper link geometry and directly constrained link velocities would lower the shield's own floor (Section 4.7).

Other architectures and moving obstacles bound the generality of the result. The case that matters for deployment is a constraint that moves.

Finally, **separating the surviving explanations** would say something general about when demonstrations transfer between controllers. A direct comparison against model-based safe RL on this benchmark [15] would place distillation against the strongest current alternative.

6.5 Contribution of the Study

Learning safety from CBF-related supervision is not new; it has been treated theoretically [4], and learning from CBF demonstrations in simulation predates that work. Distilling a corrective signal into a vision-language-action policy is not new either [17].

What is new is the conjunction: a control-theoretic safety filter, distilled from a VLA's own shielded rollouts and evaluated with the filter removed, in the setting where runtime shielding is the current state of the art.

Three findings bear directly on how such a system would be deployed. Distilling before shielding produces the lowest collision rate of any configuration measured, at a success rate indistinguishable from shielding the base policy, so where a runtime constraint is required it is better applied to a distilled policy than to a raw one. The distilled policy also invokes the shield roughly half as often, which matters wherever motion must be predictable. And safe demonstrations are approximately twice as cheap to collect with the shield active as without it, a direct saving for any deployment gathering task-specific data.

6.6 Closing

This study asked whether a shield's corrections could be compiled into the policy's weights, and found that they can: distilled on its own shielded rollouts, $\pi_{0.5}$ became both safer and more capable with no shield at inference, and the corrections transferred only when they had been applied to states the policy itself reaches.

The practical conclusion is narrower than "safety can be learned". A runtime filter teaches a policy the regularities it observes and corrects, and those corrections amortise into weights without the per-step optimisation comparable learned-barrier methods retain at inference [10, 14], and at a fraction of their training cost. But the learned policy is only as broad as the states, hazards and geometry those corrections represent. Runtime shielding remains necessary outside that envelope. Distillation reduces dependence on the shield; it does not remove the need for a safety case.

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Appendix A

Implementation notes

Declaration of AI Tool Usage

In accordance with module guidelines, I acknowledge the use of **Claude Code** (Anthropic) as an AI assistant during the preparation of this project:

- **Code Implementation & Experiments:** Claude Code was utilized to assist in structuring experimental pipelines, refactoring code modules, debugging runtime errors, and setting up reproduction scripts. All core logic, model choices, and experimental designs were independently directed and verified.
- **Report Structure & Formatting:** Claude Code was consulted for high-level outline recommendations, Section/Chapter organizational strategies, and LaTeX structural formatting. All narrative text, analytical discussion, and interpretation of results remain entirely my own original work.